
AI for Farmers: A YOLO and Language Model-Based Plant Disease Detection and Advisory System

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Abstract:

Plant diseases threaten global food security, causing up to 40% of crop yield losses [1]. Smallholder farmers are particularly vulnerable due to limited access to diagnostic and advisory services. This study presents an open-source prototype that integrates YOLOv8 for image-based plant disease detection [2] with a retrieval-augmented advisory module built using LangChain [3]. Agricultural extension manuals and scientific literature [4]–[18] were curated and indexed, enabling the system to generate farmer-friendly prevention and treatment guidance accurately and accessible.

The system was trained on the Crop Pest and Disease Detection Dataset (CCMT) [19], covering four crops (*cashew*, *cassava*, *maize*, and *tomato*) across 20 disease classes. At epoch 50, YOLOv8 achieved a precision of 0.852, recall of 0.862, mAP@50 of 0.894, and mAP@50–95 of 0.745. Strongest performance was observed on distinctive diseases such as *Cassava mosaic* and *Cashew red rust*, while weaker results occurred on subtle conditions such as *Cassava brown spot* and *Maize leaf spot*.

Compared with existing platforms, the proposed system contributes openness, offline operation, and a literature-based advisory workflow. These features address gaps in transparency, adaptability, and sustainability, demonstrating the potential of AI-enabled tools to strengthen smallholder resilience and advance the One Health agenda.

Keywords: YOLOv8, LangChain, LLMs, Advisory Systems, Plant disease detection

1. Introduction

Plant diseases are a major threat to agricultural productivity and food security, with the Food and Agriculture Organization (FAO) estimating that up to 40% of global crop yields are lost annually to pests and pathogens, equivalent to over USD 220 billion in economic losses [1]. Climate change further intensifies these risks by creating favorable conditions for new and more severe outbreaks [20]. For smallholder farmers in Africa and other resource-constrained regions, such losses undermine household income, exacerbate malnutrition, and increase vulnerability. Farmers often resort to pesticide use, which can damage ecosystems and pose health risks [21], underscoring the need for sustainable solutions framed within the One Health perspective [22–24].

Conventional plant disease detection relies on agricultural extension services, laboratory testing, or expert knowledge, all of which are costly, time-consuming, and often inaccessible to smallholders. These limitations delay intervention and reduce the effectiveness of control measures. Recent advances in artificial intelligence (AI),

particularly deep learning and mobile-based applications, present opportunities to improve timely diagnosis and advisory support. However, most existing tools remain limited in crop scope, language accessibility, and often depend on stable internet connectivity, restricting their use in rural areas [28]–[30].

This study introduces “AI-ε” an AI-driven tool that puts indigenous farming communities at the center of climate adaptation that integrates YOLOv8 [2] for plant disease detection with a large language model (LLM)-based advisory pipeline for treatment and prevention. By combining robust detection with accessible, evidence-based recommendations, the system aims to empower smallholder farmers with timely knowledge. Our approach positions plant health as a critical yet underexplored dimension of One Health, where safeguarding crops directly supports environmental sustainability, animal well-being, and human nutrition [23],[25].

2. Literature Review

Plant diseases remain a critical challenge to global agriculture, accounting for significant yield losses and threatening food security. The FAO estimates that up to 40% of global crop yields are lost annually to pests and diseases [1]. Savary et al. reported similar findings across five staple crops (wheat, rice, maize, potato, and soybean) with yield losses ranging from 17% to 30%, with the most severe impacts in food-insecure regions [26]. These burdens are further compounded by climate change, which reshapes the suitability of major food crops and alters the distribution of pests and pathogens [20].

This is not only an agricultural issue but also a health concern: poor crop management contributes to malnutrition, weakened immune systems, and socio-economic instability due to scarce access to food. Viewed through a One Health lens [22], crop management plays a crucial role in preventing malnutrition. Poor soil and plant health, caused by erosion, pests, disease, or inappropriate agrochemical use, reduces both yield and nutrient density in staple crops, depriving communities of essential vitamins, minerals, and proteins. Conversely, integrating plant health practices such as biofortification, soil fertility management, and agroforestry supports human, animal, and ecosystem well-being [23], [25]. As defined in the Quadripartite *One Health Joint Plan of Action (2022–2026)*, “the health of humans, domestic and wild animals, plants, and the wider environment ... are closely linked and inter-dependent,” requiring collaboration across sectors to ensure safe and nutritious food [24].

Disease control remains central to plant health. In the absence of timely and reliable diagnosis, farmers frequently rely on pesticide use as a default strategy. While effective in the short term, excessive pesticide application can lead to soil and water contamination, biodiversity loss, and health risks, raising concerns for long-term sustainability [21]. This highlights the urgency of developing alternative approaches that emphasize prevention, timely detection, and evidence-based advisory systems.

Artificial intelligence (AI), particularly deep learning, has shown considerable promise in advancing automated plant disease detection. The PlantVillage dataset [27] has been pivotal in enabling research on large-scale image-based diagnostics. Mohanty et al. demonstrated the potential of Convolutional Neural Networks (CNNs) trained on

PlantVillage, achieving high classification accuracy across multiple crops [28]. Building on this work, Ramcharan et al. applied deep learning to cassava diseases [29] and later developed mobile-based diagnostic models [30]. More recently, researchers have explored object detection frameworks such as YOLOv4 [31] and broader deep learning pipelines for crop disease detection [32]. Other frontier approaches combine CNNs with large language models (LLMs), offering the potential for both detection and context-aware advisory support [33]. However, most existing studies remain confined to controlled datasets, limiting their scalability in real-world farming conditions.

AI can drive climate innovation within a One Health framework by integrating complex data on ecosystems, agriculture, and human and animal health to predict risks and optimize interventions. Machine learning enables precision agriculture to cut emissions and malnutrition, supports early warning of zoonotic diseases, and improves biodiversity monitoring. By linking environmental, plant, and public health data, AI helps design resilient food systems and low-carbon strategies. This cross-sectoral intelligence aligns with the FAO–WHO–WOAH–UNEP One Health vision, where human, animal, plant, and ecosystem health are interdependent and require coordinated, science-based action for climate-resilient, sustainable development [22]–[24], [37].

Parallel to these research efforts, mobile applications have emerged as practical tools for farmers. PlantVillage Nuru provides real-time, offline diagnosis of cassava viral diseases and has achieved near-expert accuracy in field evaluations [34]. A key strength of Nuru is its offline functionality, but its scope is limited to cassava and a narrow set of diseases. Plantix, developed by PEAT GmbH, uses smartphone-based image recognition to identify pests and diseases across multiple crops and has been tested in partnership with CABI in India [35], [36]. Unlike Nuru, however, Plantix requires internet connectivity for cloud-based inference and has been critiqued for its commercial orientation. Both applications tend to frame plant health in isolation, overlooking its broader connections to ecosystems, human well-being, and the One Health perspective.

Taken together, the literature underscores three gaps. First, while the global burden of crop diseases has been well-documented [1], [20], [26], scalable field-ready solutions remain scarce. Second, although CNN-based approaches achieve strong results on benchmark datasets [27]–[32], real-world deployment lags behind. Finally, existing mobile advisory systems [34]–[36] provide valuable services but remain constrained by connectivity, crop coverage, and limited integration with sustainability goals. This study addresses these gaps by introducing an open-source, offline-friendly prototype that integrates YOLOv8 for multi-crop disease detection with a retrieval-based advisory system grounded in validated agricultural knowledge.

3. Methodology

The study employed a two-phase methodology to develop an open-source prototype for plant disease detection and advisory delivery. The first phase focused on image-based disease detection using a deep learning model, while the second phase implemented a retrieval-based advisory system with a RAG-inspired workflow to provide farmer-friendly prevention and treatment advice.

3.1 Dataset

The study utilized the publicly available Crop Pest and Disease Detection Dataset (CCMT) [19], created by researchers from the University of Energy and Natural Resources (UENR), in Ghana. The dataset includes images of four major crops: cashew, cassava, maize, and tomato, collected from farms across Ghana. All images were validated by expert plant virologists and pathologists to ensure reliability.

A subset of high-quality images was selected, excluding two pest classes (Grasshopper and Fall Armyworm) because they depicted insects rather than symptomatic plant tissue. To optimize annotation effort, only images with clearly visible disease symptoms were retained. All images were uploaded to Roboflow [38], where bounding boxes were manually drawn around visible symptoms, a process that required significant annotation effort to ensure accuracy. To maintain class balance across the dataset, samples were randomly selected from each disease category so that underrepresented classes were not overshadowed by majority classes.

The final dataset comprised 13,519 raw images across 20 plant disease classes. Roboflow was used to split the dataset into training (92%), validation (4%), and testing (4%) subsets. Data augmentation was applied only to the training set, generating ~3 variants per image through flipping, random cropping, and photometric adjustments ($\pm 10\%$ brightness, saturation, exposure). After augmentation, the training set expanded to ~32,094 images, while the validation and testing sets contained 1,414 and 1,407 images, respectively. All images were auto-oriented and resized to 640×640 pixels to meet YOLO input requirements.

3.2 YOLOv8 Detection Module

The detection module was implemented using YOLOv8m, a lightweight object detection model optimized for real-time inference. The model was initialized with pre-trained weights ([yolov8m.pt](#)) and fine-tuned on the annotated dataset using the Ultralytics YOLOv8 framework [2].

Training was conducted for 60 epochs with an image size of 640×640 pixels and a batch size of 16, chosen based on GPU memory constraints. The model learned to detect bounding boxes and classify 20 plant disease categories. Its efficiency and modularity made it well-suited for high-variation agricultural imagery.

3.3 Hardware Specification

All model training and evaluation tasks for the YOLOv8-based image detection system were performed on a machine equipped with an NVIDIA Tesla T4 GPU with 15 GB of VRAM, using CUDA version 12.4 and driver version 550.54.15. The GPU operated under default compute settings with no concurrent GPU-intensive processes during training.

3.4 Advisory System Design

The advisory module was developed to provide farmers with reliable, safe, and accessible prevention and treatment information following disease detection. It was implemented as a retrieval-based advisory system with a RAG-inspired workflow, combining information retrieval with large language model (LLM) summarization. The design transforms

technical agricultural knowledge into farmer-oriented guidance while maintaining accuracy, safety, and predictability of outputs. The workflow is divided into an offline preprocessing phase and a runtime retrieval phase.

In the offline preprocessing phase, agricultural resources such as extension manuals and scientific articles were curated and stored in a structured disease library in PDF format. Each document was ingested using a PDF loader, segmented into overlapping text chunks with a recursive splitter, and converted into vector embeddings using OpenAI's embedding model [39]. These preprocessing and retrieval workflows were implemented using the LangChain framework [3], which provided standardized components for document loading, text splitting, embedding generation, and similarity-based retrieval from the local vector database. The resulting embeddings were indexed in a local database to enable efficient retrieval of the most relevant passages for each crop-disease pair.

For each disease entry, separate prompts were designed for prevention and treatment advice. Prevention prompts restricted outputs to short, actionable non-chemical measures such as sanitation, pruning, resistant varieties, crop rotation, and biological controls. Treatment prompts required outputs to follow a fixed structure with three sections: (1) non-chemical measures, (2) chemical options as a last resort, and (3) conditions under which farmers should seek further help. To ensure safety, constraints required at least two non-chemical measures before any chemical option, restricted chemical advice to active ingredients rather than brand names, and excluded banned or highly hazardous pesticides such as carbendazim, paraquat, and monocrotophos. Chemical recommendations also specified clear application triggers, repeat intervals, and a safety reminder to follow label instructions and use personal protective equipment.

All outputs were compiled into CSV format, forming a structured disease catalog for use during runtime. The content was then passed through a post-processing script that standardized phrasing, simplified technical terminology, and ensured consistency across diseases. By generating and storing this content offline, the system reduced the cost and connectivity demands of real-time LLM queries, making it suitable for low-resource farming contexts.

In the runtime phase, once YOLOv8 identified a disease class from an uploaded plant image, the detection output was used to query the knowledge base. The retrieval module matched the identified disease with the corresponding entry in the catalog and delivered prevention and treatment guidance through the application interface. This design ensured that farmers received accurate, actionable, and context-appropriate recommendations in real time, without requiring continuous internet connectivity or high computational resources.

3.5 System Architecture

The integrated prototype combines the detection and advisory modules into a unified pipeline designed for usability in resource-constrained farming environments (Figure 3). The architecture consists of two complementary phases: (1) offline preprocessing, where

agricultural resources are summarized by a large language model (LLM) into a structured knowledge base using LangChain [3] and OpenAI's embedding model [39]; and (2) runtime operation, where YOLOv8 [2] detects plant diseases from uploaded images and the retrieval module delivers farmer-friendly prevention and treatment guidance. This separation ensures that the system remains lightweight, scalable, and practical for deployment while allowing expansion to additional crops and regions.

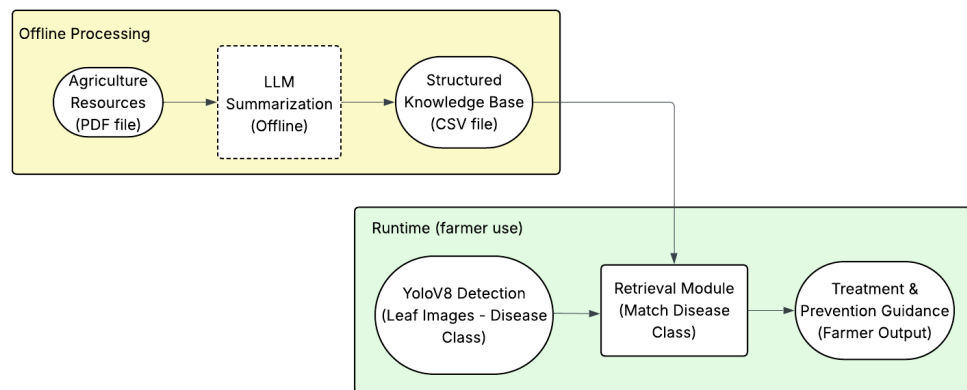


Figure 3. System architecture of the AI for Farmers prototype. The pipeline has two phases: (1) offline preprocessing, where agricultural resources (e.g., manuals, PDFs) are summarized by an LLM into a structured knowledge base; and (2) runtime operation, where YOLOv8 detects plant diseases from farmer-uploaded images and the retrieval module delivers farmer-friendly prevention and treatment guidance.

3.6 Evaluation

The system was evaluated along two dimensions: technical performance and practical usability. For the disease detection module, performance was quantified using mean Average Precision (mAP), precision, recall and F1 score on the held-out test set. These metrics provided a robust measure of the model's ability to accurately localize and classify disease symptoms across 20 crop-disease classes.

For the advisory module, recommendations were generated from curated agricultural extension manuals and scientific reports [4]–[18] ingested into the system as PDF inputs. The evaluation therefore focused on verifying that outputs complied with predefined design constraints, such as prioritizing non-chemical measures, excluding banned pesticides, and presenting advice in clear, farmer-friendly language. Although expert validation was not conducted at this stage, the reliance on trusted extension resources and the application of rule-based safeguards contributed to the clarity, safety, and practical relevance of the advisory content. Future work will include validation of advisory outputs by agricultural experts and field testing with farmers.

4. Results

4.1 Detection Performance

The YOLOv8 model was trained for 80 epochs on the plant disease dataset. However, inspection of training and validation loss curves indicated signs of overfitting beyond epoch 50 (see Figure 4.1a). Specifically, validation loss plateaued while training loss continued to decline, suggesting reduced generalization. To avoid reporting overfitted results, performance is presented at epoch 50, which was identified as the optimal trade-off point between accuracy and generalization.

At epoch 50, the model achieved an overall precision of 0.852, recall of 0.862, mAP@50 of 0.894, and mAP@50-95 of 0.745 (see Figure 4.1b).

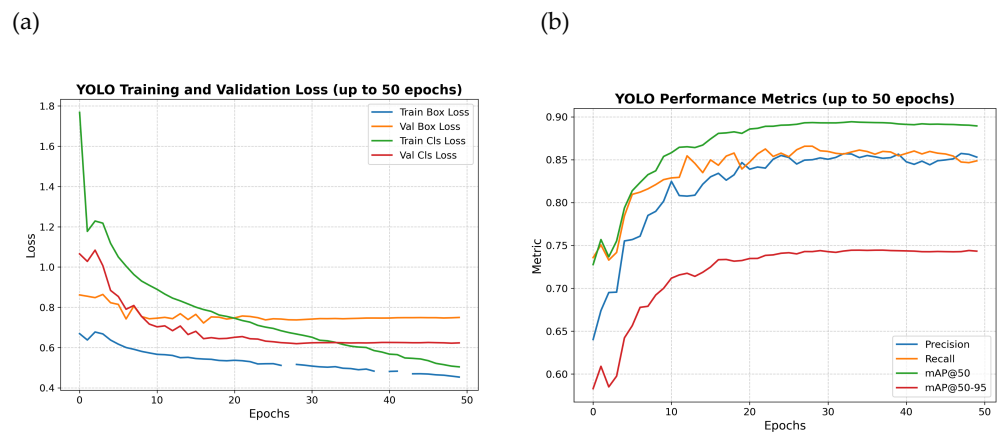


Figure 4.1: (a) YOLO training and validation loss curves (up to 50 epochs) (right). (b) YOLO performance metrics across epochs (left).

Class-wise results are presented in **Table 4.1**. The model performed best on *Cassava mosaic* (mAP@50-95 = 0.980, F1 = 0.994), *Cashew red rust* (0.970, F1 = 0.978), and *Cassava green mite* (0.970, F1 = 0.945). In contrast, the weakest performance was recorded for *Cassava brown spot* (0.240, F1 = 0.591), *Cassava bacterial blight* (0.348, F1 = 0.702), and *Maize leaf spot* (0.411, F1 = 0.619). These disparities indicate that the model generalizes effectively to diseases with clear visual signatures, but struggles with classes where symptoms are overlapping, subtle, or easily confused. A complete class-wise breakdown is provided in **Appendix A (Table A.1)**.

Table 4.1 Class-wise performance of YOLOv8 at epoch 50, showing precision, recall, mAP@50, mAP@50-95, and F1 score for selected crop-disease classes.

Crop / Disease	Precision (P)	Recall (R)	mAP@50	mAP@50-95	F1
Overall (all classes)	0.852	0.862	0.894	0.745	0.857
<i>Cassava mosaic</i>	0.989	1	0.995	0.98	0.994
<i>Cashew red rust</i>	0.976	0.981	0.989	0.97	0.978
<i>Cassava green mite</i>	0.987	0.907	0.987	0.97	0.945

<i>Cassava brown spot</i>	0.669	0.529	0.611	0.24	0.591
<i>Cassava bacterial blight</i>	0.709	0.695	0.722	0.348	0.702
<i>Maize leaf spot</i>	0.749	0.52	0.617	0.411	0.614

Representative detection samples are illustrated in Figure 4.2, while a complete set of detection outputs covering all 20 crop–disease classes is provided in **Appendix B (Figure B.1)**. The model successfully localized and classified symptoms across all four crops. For cashew, YOLOv8 correctly identified mosaic symptoms with high confidence (0.99). In cassava, the model detected leaf miner damage, with bounding boxes highlighting multiple affected regions. For maize, foliar lesions of leaf blight were localized with confidence of 0.80, while tomato detection accurately identified *Septoria* leaf spot at 1.00 confidence. These examples demonstrate the system’s ability to detect both single- and multi-lesion cases, as well as to distinguish between healthy and diseased leaves. They complement the quantitative results presented in Table 4.1 by visually confirming the detection module’s capacity to generalize across different crops and diseases.

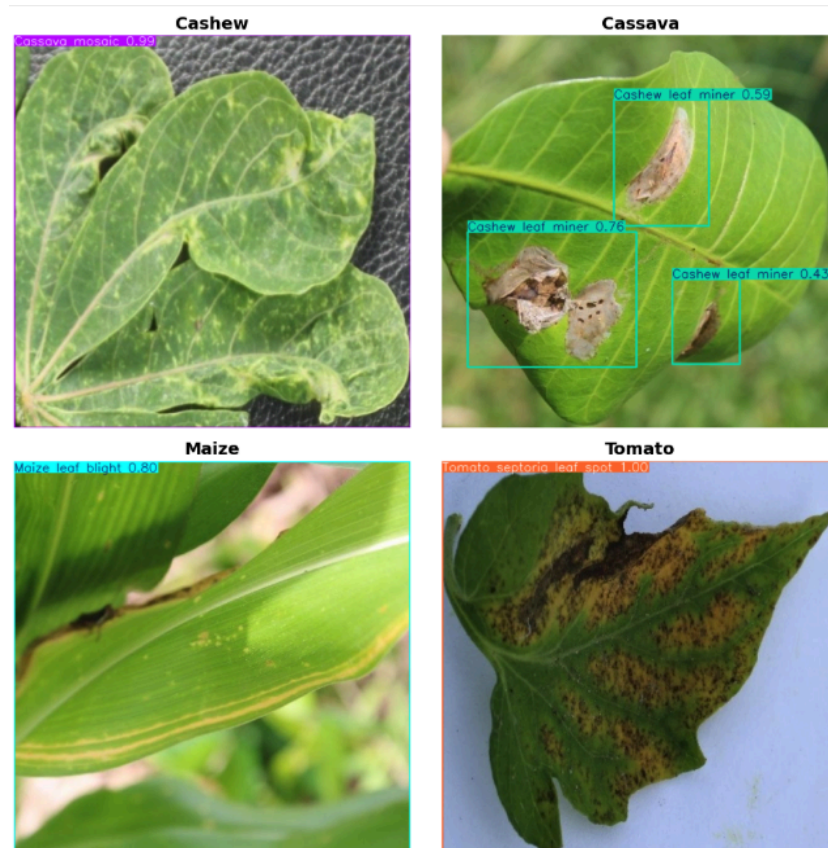


Figure 4.2. Representative YOLOv8 detection outputs for cashew, cassava, maize, and tomato in the validation set. Bounding boxes indicate predicted disease classes and confidence scores.

4.2 Advisory System Evaluation

The advisory module was evaluated against recent scientific and extension literature for each crop–disease pair. A curated set of studies provided the reference knowledge base,

including management guidelines for *Cashew gummosis* [20], *Cashew leaf miner* [21], *Cashew red rust* [22], *Cassava bacterial blight* [23], *Cassava brown spot* [24], *Cassava green mite* [25], *Cassava mosaic* [26], *Maize leaf beetle* [27], *Maize leaf blight* [28], *Maize leaf spot* [29], *Maize streak virus* [30], and tomato diseases including early blight [31], leaf curl [32], *Septoria leaf spot* [33], and *Verticillium wilt* [34].

The majority of recommendations generated by the system were consistent with these validated sources, particularly in highlighting non-chemical strategies such as resistant varieties, pruning, sanitation, and crop rotation. Representative advisory outputs are shown in Figure 4.3. For example, the system's guidance for tomato *Septoria leaf spot* presents prevention and treatment advice in farmer-friendly bullet points. Prevention emphasizes resistant varieties, pruning, and crop rotation, while treatment prioritizes non-chemical measures, with chemical fungicides included only as a last resort and accompanied by explicit safety instructions.

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=== Tomato - Septoria Leaf Spot ===
=== Prevention ===
- Plant resistant tomato varieties to Septoria leaf spot.
- Space tomato plants properly for good air circulation.
- Remove and destroy infected leaves as soon as they are noticed.
- Mulch around tomato plants to prevent soil splashing onto leaves.
- Rotate tomato crops with non-related plants to break the disease cycle.
- Inspect tomato plants weekly for early signs of Septoria leaf spot.

=== Treatment ===
Non-chemical:
- Remove infected leaves and stems promptly.
- Ensure good spacing between plants for airflow.

Chemical (last option):
- Copper fungicide – when: signs of Septoria leaf spot appear; interval:
every 7–10 days; note: Follow label instructions and wear PPE.

Seek help when:
- Disease continues to spread despite treatment efforts.

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Figure 4.3. Example advisory outputs for tomato *Septoria leaf spot*, showing prevention and treatment guidance generated by the system.

While many of the reference studies included chemical control options, the advisory module was deliberately designed to prioritize environmentally sustainable and accessible practices. Chemical measures were excluded or de-emphasized, and instead the system emphasized safe, low-cost, and culturally appropriate strategies aligned with an integrated pest management (IPM) approach.

Overall, the advisory module demonstrated strong alignment with validated agricultural knowledge and successfully transformed technical information into accessible, farmer-oriented guidance. Future work will include expert validation of outputs and field testing with smallholder farmers to further improve relevance and adoption.

5. Discussion

This study introduced an open-source plant disease detection and advisory system combining YOLOv8 for image-based diagnosis with a retrieval-based advisory module. The system was designed to provide smallholder farmers with accurate detection and farmer-friendly, sustainable management recommendations.

The detection results demonstrated that YOLOv8 achieved strong overall performance, with a precision of 0.852, recall of 0.862, mAP@50 of 0.894, and mAP@50-95 of 0.745 at epoch 50. High accuracies were observed for visually distinctive diseases such as *Cassava mosaic* (mAP@50-95 = 0.980), *Cashew red rust* (0.970), and *Cassava green mite* (0.970), confirming the model's effectiveness when symptoms present clear morphological patterns. By contrast, weaker performance on *Cassava brown spot* (0.240), *Cassava bacterial blight* (0.348), and *Maize leaf spot* (0.411) highlights the challenges of class imbalance and overlapping symptom expression. These findings are consistent with prior studies showing that dataset size and symptom distinctiveness are critical for robust plant disease detection.

The advisory system produced outputs that were clear, practical, and consistent with validated agricultural literature [19]–[33]. A key strength of the module is its deliberate emphasis on non-chemical, environmentally friendly measures such as pruning, sanitation, and resistant varieties, ensuring both ecological sustainability and accessibility for smallholder farmers. While this approach distinguishes the system from conventional manuals that prioritize chemical pesticides, the inclusion of optional integrated pest management (IPM)-compatible chemical measures may enhance completeness for advanced users.

Compared with existing mobile applications such as PlantVillage Nuru and Plantix, the proposed system provides three notable contributions: open-source accessibility, multi-crop coverage across four staple crops, and a retrieval-based advisory workflow that dynamically generates guidance grounded in validated literature. These features address gaps in transparency, adaptability, and environmental responsibility often associated with commercial or closed-source platforms.

Beyond accuracy, the prototype operationalizes One Health by pairing early image-based detection with advice that minimizes hazardous chemical use, protects farmer/household health, and reduces environmental contamination. Fewer sprays and clearer PPE can lower acute/chronic exposures; non-chemical controls preserve pollinators and waterways; and steadier yields support nutrition and income. Open-source, offline delivery improves access for underserved smallholders, advancing health equity. Safety is reinforced through retrieval from vetted sources and rule-based constraints on actives, with expert review planned. Together, these features position AI-enabled plant health as a lever for human, animal, and ecosystem well-being under climate stress [21–25].

Despite these contributions, several limitations must be acknowledged. The current prototype is restricted to four crops: cashew, cassava, maize, and tomato, which limits its immediate generalizability. Detection performance may vary across diseases, particularly those with visually similar symptoms, and additional validation under real-world farm conditions is still required. The advisory system, while effective in providing prevention and treatment information, remains retrieval-based with static outputs rather than an interactive, conversational advisory tool. Furthermore, multilingual and low-literacy features have not yet been integrated, though these are critical for accessibility in diverse

African farming communities. Addressing these limitations will be essential before the system can achieve broad adoption.

Another important strength of the chosen architecture is its extensibility. The advisory system was implemented using LangChain [17], providing a modular framework for retrieval-augmented generation (RAG) pipelines. This design makes it relatively straightforward to add functionalities such as multilingual support, chatbot-style farmer interactions, or links to external agricultural databases. While these features are not yet included, the system's modularity ensures that they can be incorporated with minimal structural changes, supporting future scalability and adaptability.

Finally, it is important to acknowledge that the field of object detection is rapidly evolving. While YOLOv8 was employed in this study due to its strong balance of accuracy and efficiency, newer versions such as YOLOv11 have recently been introduced. YOLOv11 demonstrates enhanced real-time inference capabilities and improved benchmark accuracy compared to YOLOv8. Future work could explore migrating the detection module to YOLOv11 to leverage these improvements, further strengthening the system's ability to support real-time, field-based plant health monitoring.

In summary, the proposed system demonstrates that integrating state-of-the-art detection with retrieval-based advisory generation can deliver reliable, clear, and environmentally responsible disease management guidance. Although further refinement and field validation are needed, this work contributes an important step toward scalable, open-source decision support tools for smallholder agriculture. Most importantly, it illustrates how AI-enabled crop protection can advance the One Health agenda by linking safer farming practices with improved human health, sustainable food production, animal well-being, and environmental resilience.

6. Conclusions

This study introduced an open-source plant disease detection and advisory system that combines YOLOv8-based image recognition with a retrieval-augmented advisory pipeline. The detection module achieved strong overall performance, particularly for diseases with distinct visual symptoms such as *Cassava mosaic* and *Cashew red rust*, while highlighting challenges in detecting subtle or overlapping conditions such as *Cassava brown spot* and *Maize leaf spot*. The advisory module provided clear, actionable, and sustainable guidance aligned with validated agricultural sources, emphasizing non-chemical measures and restricting chemical advice to last-resort options with explicit safety reminders.

Compared with existing platforms such as PlantVillage Nuru and Plantix, the proposed system offers three key contributions: open-source accessibility, offline-friendly operation, and multi-crop coverage across four staple crops. These features address barriers of connectivity, transparency, and scope that limit adoption of current tools.

Limitations include restricted crop coverage, dataset imbalance, and the absence of multilingual and farmer-centered usability features. However, the modular design,

implemented using LangChain, provides a foundation for extending the system with conversational interfaces, localized content, and future advances in detection models such as YOLOv11.

Overall, this work demonstrates that integrating advanced AI detection with retrieval-based advisory generation can deliver scalable, accessible, and environmentally responsible decision-support tools for agriculture. It also illustrates how such platforms can strengthen early detection, rapid response, and precision interventions, reducing chemical use and safeguarding yields. By integrating plant, animal, and environmental health data, the system directly supports the One Health approach, which recognizes the interconnection between human health, ecosystems, and food systems. Such AI-driven tools enhance resilience to pests, pathogens, and climate stress while fostering sustainable agriculture and food security. Ultimately, AI-enabled decision support promotes preventive, data-informed management that protects biodiversity, reduces zoonotic risks, and advances global goals for health, environment, and nutrition within an integrated One Health framework.

Supplementary Materials: https://huggingface.co/spaces/aycu2004/ai_for_farmer

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Appendix A.1. Class-Wise Detection Performance

This appendix provides the complete class-wise detection performance of the YOLOv8 model across 20 crop–disease categories. For each class, Precision, Recall, mAP@50, mAP@50–95, and F1 scores are reported. While the main text (Table 4.1) presents a selected subset of results, this appendix ensures full transparency by including all categories, highlighting where the model performed strongly (e.g., *Cassava mosaic*, *Cashew red rust*) and where detection was weaker (e.g., *Cassava brown spot*, *Maize leaf spot*).

Table A.1. Class-wise YOLOv8 detection performance across 20 crop–disease categories, showing Precision, Recall, mAP@50, mAP@50–95, and F1 scores.

Crop / Disease	Precision (P)	Recall (R)	mAP@50	mAP@50-95	F1
Overall (all classes)	0.852	0.862	0.894	0.745	0.857
<i>Cashew anthracnose</i>	0.973	0.967	0.992	0.957	0.97
<i>Cashew gumosis</i>	0.925	0.916	0.948	0.578	0.921
<i>Cashew healthy</i>	0.928	0.962	0.981	0.678	0.945
<i>Cashew leaf miner</i>	0.745	0.755	0.779	0.418	0.75
<i>Cashew red rust</i>	0.976	0.981	0.989	0.97	0.978
<i>Cassava bacterial blight</i>	0.709	0.695	0.722	0.348	0.702
<i>Cassava brown spot</i>	0.669	0.529	0.611	0.24	0.591
<i>Cassava green mite</i>	0.987	0.907	0.987	0.97	0.945
<i>Cassava healthy</i>	0.784	0.792	0.859	0.547	0.788
<i>Cassava mosaic</i>	0.989	1	0.995	0.98	0.994
<i>Maize healthy</i>	0.83	1	0.946	0.906	0.907
<i>Maize leaf beetle</i>	0.808	0.88	0.919	0.645	0.842
<i>Maize leaf blight</i>	0.934	0.941	0.961	0.874	0.938
<i>Maize leaf spot</i>	0.749	0.52	0.617	0.411	0.614
<i>Maize streak virus</i>	0.91	0.887	0.923	0.859	0.898
<i>Tomato healthy</i>	0.894	1	0.971	0.958	0.944
<i>Tomato leaf blight</i>	0.785	0.747	0.806	0.732	0.766
<i>Tomato leaf curl</i>	0.792	0.886	0.945	0.915	0.836
<i>Tomato septoria leaf spot</i>	0.839	0.941	0.96	0.948	0.887
<i>Tomato verticillium wilt</i>	0.811	0.934	0.965	0.955	0.868

Appendix B.1. Complete YOLOv8 Detection Outputs

This appendix provides the complete detection outputs of the YOLOv8 model across 20 crop–disease categories in the validation set. Each image shows bounding boxes with predicted disease classes and confidence scores. While the main text (Figure 4.2) presents representative samples from cashew, cassava, maize, and tomato, this appendix ensures full transparency by including outputs for all categories. The outputs demonstrate the model’s ability to detect single and multiple lesions, distinguish between diseased and healthy samples, and highlight areas of both high and low confidence across classes.

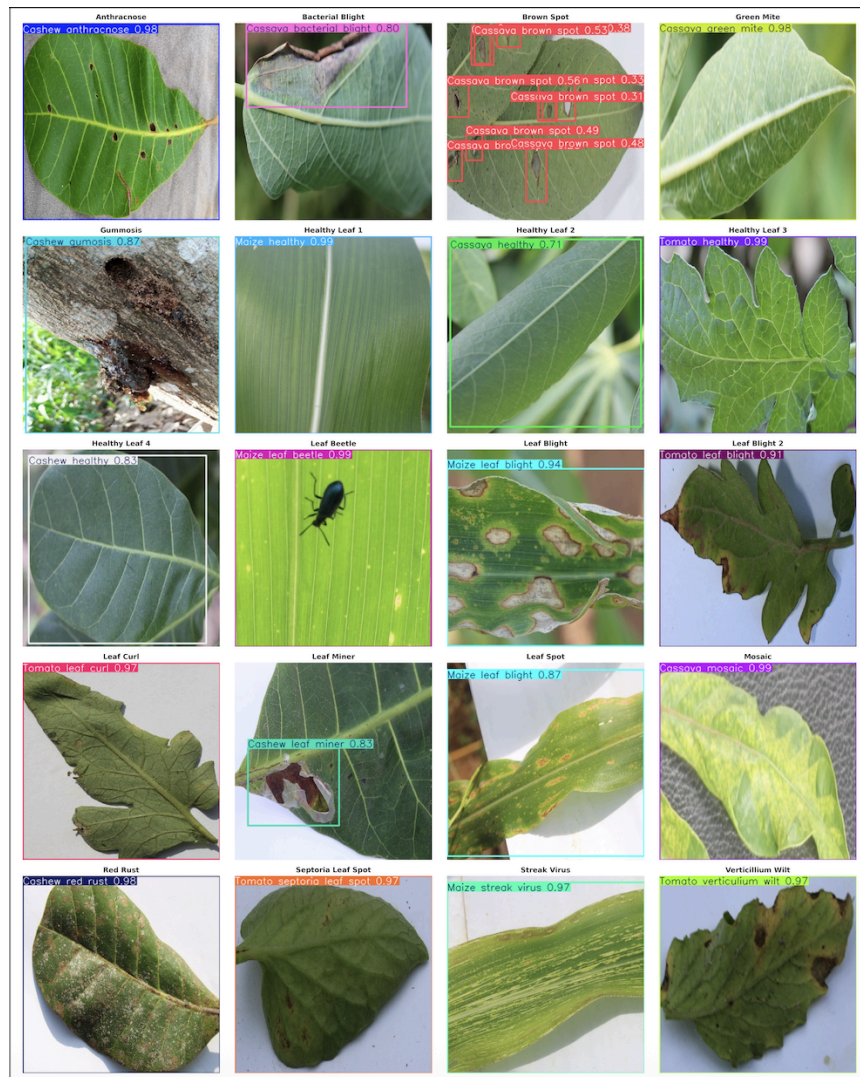


Figure B.1. Complete YOLOv8 detection outputs across 20 crop–disease categories in the validation set, with bounding boxes showing predicted disease classes and confidence scores.